MPG No. 4.3.11 m' [Type III, monoclinic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & T_y & 0 \end{bmatrix}$$

$$T_y = \mathbb{T}_y^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ T_{xy} & 0 & T_{yz} \\ 0 & T_{yz} & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{T}_{xy}^{(1)}$$
$$T_{yz} = \mathbb{T}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & T_{yz} \\ 0 & -T_{yz} & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}_z^{(1)}$$
$$T_{yz} = \mathbb{M}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & T_{xxy} & 0 \\ 0 & T_{yyy} & 0 \\ 0 & T_{zzy} & 0 \\ T_{yzx} & 0 & T_{zzy} \\ 0 & T_{yzx} & 0 \\ T_{xxy} & 0 & T_{yzx} \end{bmatrix}$$

$$T_{xxy} = \mathbb{T}_y^{(1)} - \mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\beta(1)}$$
$$T_{yyy} = 3\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)}$$
$$T_{zzy} = \mathbb{T}_y^{(1)} - \mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\beta(1)}$$
$$T_{yzx} = \mathbb{T}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & T_{zzy} & 0 \\ T_{yzx} & 0 & -\frac{T_{zzy}}{2} \\ 0 & T_{zxy} & 0 \\ -\frac{T_{xxy}}{2} & 0 & -T_{yzx} - T_{zxy} \end{bmatrix}$$

$$T_{xxy} = -2\mathbb{M}_{xz}^{(1)} + 2\mathbb{T}_y^{(2)}$$
$$T_{zzy} = 2\mathbb{M}_{xz}^{(1)} + 2\mathbb{T}_y^{(2)}$$
$$T_{yzx} = 3\mathbb{M}_u^{(1)} + \mathbb{M}_v^{(1)}$$
$$\frac{T_{zzy}}{2} = \mathbb{M}_{xz}^{(1)} + \mathbb{T}_y^{(2)}$$
$$T_{zxy} = -3\mathbb{M}_u^{(1)} + \mathbb{M}_v^{(1)}$$
$$\frac{T_{xxy}}{2} = -\mathbb{M}_{xz}^{(1)} + \mathbb{T}_y^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & 0 & T_{yzz} \\ 0 & T_{zxy} & 0 \\ T_{xyx} & 0 & T_{xyz} \end{bmatrix}$$

$$\begin{aligned}
T_{yzx} &= \mathbb{M}_{0}^{(1)} - \mathbb{M}_{u}^{(2)} + \mathbb{M}_{v}^{(2)} \\
T_{yzz} &= \mathbb{M}_{xz}^{(2)} - \mathbb{T}_{y}^{(3)} \\
T_{zxy} &= \mathbb{M}_{0}^{(1)} - \mathbb{M}_{u}^{(2)} - \mathbb{M}_{v}^{(2)} \\
T_{xyx} &= \mathbb{M}_{xz}^{(2)} + \mathbb{T}_{y}^{(3)} \\
T_{xyz} &= \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{u}^{(2)}
\end{aligned}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & T_{xxxy} \\ 0 & 0 & 0 & T_{yyyz} & 0 & T_{yyxy} \\ 0 & 0 & 0 & T_{zzyz} & 0 & T_{zzxy} \\ T_{xxyz} & T_{yyyz} & T_{zzyz} & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{yzzx} & 0 & T_{zxxy} \\ T_{xxxy} & T_{yyxy} & T_{zzxy} & 0 & T_{zxxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxyz} &= 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} \\
T_{xxxy} &= \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} \\
T_{yyyz} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} \\
T_{yyxy} &= -\mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} \\
T_{zzyz} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} \\
T_{zzxy} &= 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} \\
T_{yzzx} &= 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} \\
T_{zxxy} &= 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)}
\end{aligned}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & T_{xxxy} \\ 0 & 0 & 0 & T_{yyyz} & 0 & T_{yyxy} \\ 0 & 0 & 0 & T_{zzyz} & 0 & T_{zzxy} \\ -T_{xxyz} & -T_{yyyz} & -T_{zzyz} & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & -T_{yzzx} & 0 & T_{zxxy} \\ -T_{xxxy} & -T_{yyxy} & -T_{zzxy} & 0 & -T_{zxxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxyz} &= -4\mathbb{M}_{x}^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{xxxy} &= 2\mathbb{M}_{z}^{(1)} - 2\mathbb{M}_{z}^{\alpha(1)} + 2\mathbb{M}_{z}^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{yyyz} &= 2\mathbb{M}_{x}^{(1)} - 2\mathbb{M}_{x}^{\alpha(1)} + 2\mathbb{M}_{x}^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yyxy} &= -2\mathbb{M}_{z}^{(1)} + 2\mathbb{M}_{z}^{\alpha(1)} + 2\mathbb{M}_{z}^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{zzyz} &= -2\mathbb{M}_{x}^{(1)} + 2\mathbb{M}_{x}^{\alpha(1)} + 2\mathbb{M}_{x}^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{zzxy} &= -4\mathbb{M}_{z}^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{yzzx} &= -\mathbb{M}_{z}^{(1)} - 4\mathbb{M}_{z}^{\alpha(1)} \\
T_{zxxy} &= -\mathbb{M}_{x}^{(1)} - 4\mathbb{M}_{x}^{\alpha(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ T_{yzzx} & 0 & T_{zxxy} \\ 0 & T_{zxxy} & 0 \end{bmatrix}$$

$$T_{yzzx} = 2\mathbb{T}^{(4)}_{xy}$$
$$T_{zxxy} = 2\mathbb{T}^{(4)}_{yz}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ -T_{yzzx} & 0 & T_{zxxy} \\ 0 & -T_{zxxy} & 0 \end{bmatrix}$$

$$T_{yzzx} = -\mathbb{M}^{(2)}_{z}$$
$$T_{zxxy} = -\mathbb{M}^{(2)}_{x}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & 0 & T_{xxxy} \\ T_{yyyz} & 0 & T_{yyxy} \\ T_{zzyz} & 0 & T_{zzxy} \\ 0 & T_{yzzx} & 0 \\ T_{zxyz} & 0 & T_{zxxy} \\ 0 & T_{xyzx} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}^{(3)}_{x} + 2\mathbb{M}^{(4)}_{x} + 2\mathbb{M}^{\alpha(2)}_{x}$$
$$T_{xxxy} = \mathbb{M}^{(3)}_{z} - \mathbb{M}^{\alpha(2)}_{z} + \mathbb{M}^{\beta(2)}_{z} - 2\mathbb{T}^{(5)}_{xy}$$
$$T_{yyyz} = \mathbb{M}^{(3)}_{x} - \mathbb{M}^{\alpha(2)}_{x} + \mathbb{M}^{\beta(2)}_{x} - 2\mathbb{T}^{(5)}_{yz}$$
$$T_{yyxy} = \mathbb{M}^{(3)}_{z} - \mathbb{M}^{\alpha(2)}_{z} - \mathbb{M}^{\beta(2)}_{z} + 2\mathbb{T}^{(5)}_{xy}$$
$$T_{zzyz} = \mathbb{M}^{(3)}_{x} - \mathbb{M}^{\alpha(2)}_{x} - \mathbb{M}^{\beta(2)}_{x} + 2\mathbb{T}^{(5)}_{yz}$$
$$T_{zzxy} = \mathbb{M}^{(3)}_{z} + 2\mathbb{M}^{(4)}_{z} + 2\mathbb{M}^{\alpha(2)}_{z}$$
$$T_{yzzx} = \mathbb{M}^{(4)}_{z} - \mathbb{M}^{\alpha(2)}_{z} - \mathbb{M}^{\beta(2)}_{z} - \mathbb{T}^{(5)}_{xy}$$
$$T_{zxyz} = \mathbb{M}^{(4)}_{z} - \mathbb{M}^{\alpha(2)}_{z} + \mathbb{M}^{\beta(2)}_{z} + \mathbb{T}^{(5)}_{xy}$$
$$T_{zxxy} = \mathbb{M}^{(4)}_{x} - \mathbb{M}^{\alpha(2)}_{x} - \mathbb{M}^{\beta(2)}_{x} - \mathbb{T}^{(5)}_{yz}$$
$$T_{xyzx} = \mathbb{M}^{(4)}_{x} - \mathbb{M}^{\alpha(2)}_{x} + \mathbb{M}^{\beta(2)}_{x} + \mathbb{T}^{(5)}_{yz}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & T_{zxxy} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & T_{xyzx} & 0 \end{bmatrix}$$

$$T_{yzxx} = \mathbb{M}^{(5)}_{x} + 2\mathbb{M}^{(6)}_{x} + 2\mathbb{M}^{\alpha(3)}_{x}$$
$$T_{yzyy} = \mathbb{M}^{(5)}_{x} - \mathbb{M}^{\alpha(3)}_{x} + \mathbb{M}^{\beta(3)}_{x} - 2\mathbb{T}^{(6)}_{yz}$$
$$T_{yzzz} = \mathbb{M}^{(5)}_{x} - \mathbb{M}^{\alpha(3)}_{x} - \mathbb{M}^{\beta(3)}_{x} + 2\mathbb{T}^{(6)}_{yz}$$
$$T_{yzzx} = \mathbb{M}^{(6)}_{z} - \mathbb{M}^{\alpha(3)}_{z} + \mathbb{M}^{\beta(3)}_{z} + \mathbb{T}^{(6)}_{xy}$$
$$T_{zxyz} = \mathbb{M}^{(6)}_{z} - \mathbb{M}^{\alpha(3)}_{z} - \mathbb{M}^{\beta(3)}_{z} - \mathbb{T}^{(6)}_{xy}$$
$$T_{zxxy} = \mathbb{M}^{(6)}_{x} - \mathbb{M}^{\alpha(3)}_{x} + \mathbb{M}^{\beta(3)}_{x} + \mathbb{T}^{(6)}_{yz}$$
$$T_{xyxx} = \mathbb{M}^{(5)}_{z} - \mathbb{M}^{\alpha(3)}_{z} + \mathbb{M}^{\beta(3)}_{z} - 2\mathbb{T}^{(6)}_{xy}$$
$$T_{xyyy} = \mathbb{M}^{(5)}_{z} - \mathbb{M}^{\alpha(3)}_{z} - \mathbb{M}^{\beta(3)}_{z} + 2\mathbb{T}^{(6)}_{xy}$$
$$T_{xyzz} = \mathbb{M}^{(5)}_{z} + 2\mathbb{M}^{(6)}_{z} + 2\mathbb{M}^{\alpha(3)}_{z}$$
$$T_{xyzx} = \mathbb{M}^{(6)}_{x} - \mathbb{M}^{\alpha(3)}_{x} - \mathbb{M}^{\beta(3)}_{x} - \mathbb{T}^{(6)}_{yz}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & T_{xxxy} & T_{xxzy} & 0 & T_{xxyx} \\ 0 & 0 & 0 & T_{yyyz} & 0 & T_{yyxy} & T_{yyzy} & 0 & T_{yyyx} \\ 0 & 0 & 0 & T_{zzyz} & 0 & T_{zzxy} & T_{zzzy} & 0 & T_{zzyx} \\ T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & T_{zxxy} & T_{zxzy} & 0 & T_{zxyx} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{xxxy} = 2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$

$$T_{xxzy} = -\mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{(4)} - 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{xxyx} = 2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$

$$T_{yyyz} = 2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{yyxy} = -2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$

$$T_{yyzy} = 2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{yyyx} = -2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$

$$T_{zzyz} = -2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{zzzy} = -2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzyx} = -\mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{(4)} - 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{yzxx} = 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{yzyy} = -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{yzzz} = 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{yzzx} = -\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$

$$T_{yzxz} = -\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$

$$T_{zxyz} = \mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$

$$T_{zxxy} = -\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$

$$T_{zxzy} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$

$$T_{zxyx} = -\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)}$$

$$T_{xyxx} = -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} - 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$

$$T_{xyyy} = 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)} - 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$

$$T_{xyzz} = 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} - \mathbb{T}_{xy}^{(3)}$$

$$T_{xyzx} = \mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)}$$

$$T_{xyxz} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & T_{zxxy} & T_{zxzy} & 0 & T_{zxyx} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0 \end{bmatrix}$$

$$
\begin{aligned}
T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\
T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzx} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\
T_{yzxz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\
T_{zxyz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\
T_{zxxy} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{zxzy} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\
T_{zxyx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)} \\
T_{xyyy} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)} \\
T_{xyzz} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)} \\
T_{xyzx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\
T_{xyxz} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & T_{xxxy} & 0 \\
T_{yyyx} & 0 & T_{yyyz} \\
0 & T_{zzzy} & 0 \\
0 & T_{yyzy} & 0 \\
0 & T_{zzxy} & 0 \\
T_{xxyx} & 0 & T_{xxyz} \\
T_{yzzx} & 0 & T_{yzzz} \\
0 & T_{zxxy} & 0 \\
0 & T_{xyyy} & 0 \\
-\frac{T_{yyyz}}{2} + \frac{T_{yyzy}}{2} + T_{zxxy} & 0 & \frac{T_{xxxy}}{2} - \frac{T_{xxyx}}{2} + T_{xyyy} - T_{yyyx} + T_{zzxy}
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxxy} &= \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} \\
T_{yyyx} &= -\mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} \\
T_{yyyz} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{zzzy} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyzy} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzxy} &= 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{xxyx} &= \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)} \\
T_{xxyz} &= 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yzzx} &= 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)} \\
T_{yzzz} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\
T_{zxxy} &= 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{xyyy} &= -\mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}
\end{aligned}
$$